



Planet Profit ALDI Royalty Scheme

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Table of Contents

<i>Abstract</i>	3
<i>Infrastructure Analysis</i>	4-5
<i>B2C and B2B Analysis</i>	6-8
<i>Business Intelligence</i>	8-10
<i>GDPR and Privacy</i>	10-12
<i>Ethics Canvas and Analysis</i>	13-15
<i>Business Model Canvas</i>	16-18
<i>References</i>	19-20

Abstract

We are known as Planet Profit and the company we decided to analyse is Aldi. Aldi is a budget-focused grocery retailer and a competitor for other retailers such as Lidl, Tesco, Dunnes and SuperValu. We believe Aldi is a capable retailer and we have all seen it has opened multiple stores, especially in developing areas in Ireland. One thing however that stands out to us about Aldi is its lack of royalty scheme in Ireland either through its application or loyalty points, which all of its competitors offer. Aldi lacks a form of loyalty from customers with the lack of a rewards system.

Inspired by our findings from the article *“Aldi’s Biggest Competitor, Lidl, Uses Perks and Coupons To Win Shoppers”*, our IS solution is to push forward a rewards system and personalised products just like their competitors. This involves scanning QR codes and offering incentives like free coffee to customers so that they feel valued.

The benefit this brings to ALDI is hopefully increased profits and improved customer loyalty. If customers receive special discounts like a “just for you” on a pastry, they’ll have a higher chance of buying it. This will also introduce new insights to ALDI about customers spending and more opportunities for business research and intelligence

Infrastructure Analysis

Overview

ALDI's infrastructure is a mixture of old and new technology including software that employs old central systems and new cloud apps, hardware that has in-store systems, on-premise data centres and backend servers and their networks uses private WANs that connect stores to headquarters integrated with public internet and cloud connections. Although the infrastructure is not bad by any means it pales in comparison to infrastructure used by ALDI's competitors.

Scalability and Performance

The current systems struggle to handle massive data loads that emerge from new technologies. During the global rollout of Electronic Shelf Labels (ESLs), ALDI admitted the software "could not keep up with the global rollout, which led to problems" due to the huge volume of data. This lack of scalability risks website and app crashes during busy periods and can slow down the launch of new services like the SHOP&GO cashier less stores.

Security and Compliance

Adding millions of new connected devices such as digital price tags and AI cameras creates more entry points for attackers. An ALDI Security Architect stated his biggest challenge is "the size of the infrastructure" and "the large number of potential dependencies." A security breach in this environment could lead to stolen customer data or even hackers changing prices in stores, either of which would have devastating effects on the company.

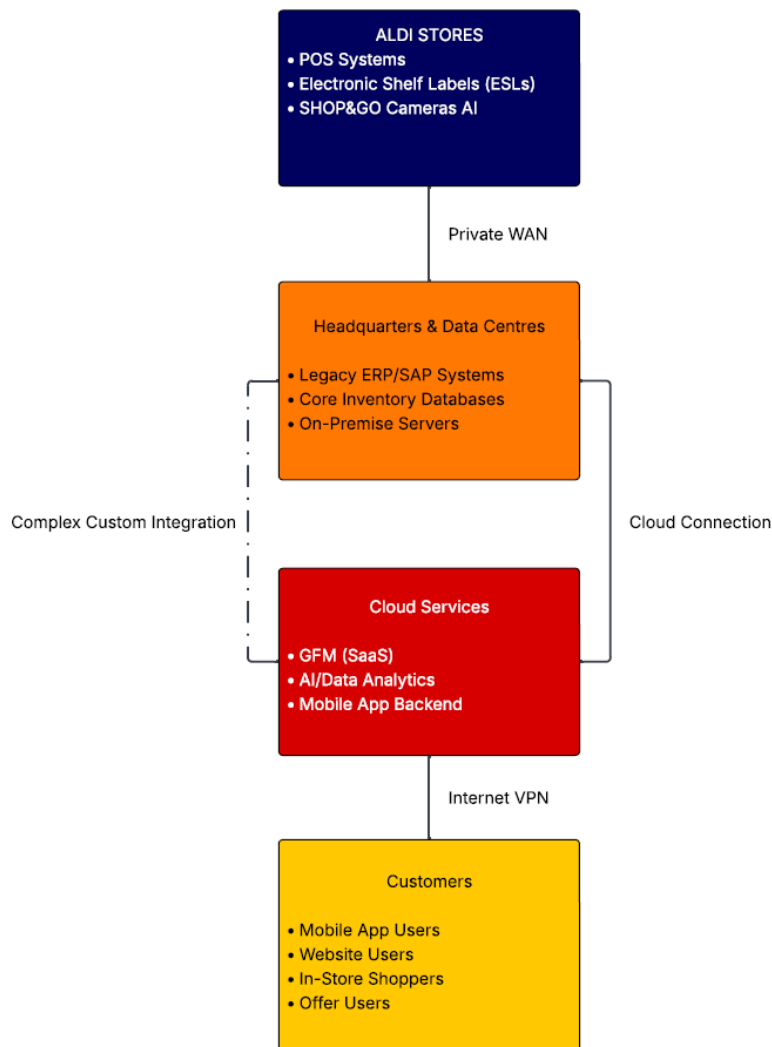
Cloud Adoption Status

While ALDI is using more cloud apps, they still rely on old "legacy" systems creating a complex hybrid environment. ALDI has officially "transitioned to more modern cloud solutions" but new apps like the facility management tool must still connect back to old central systems. This overall makes the IT landscape more complicated and fragile, slowing down innovation.

System Integration Challenges

The old and new systems are not designed to talk to each other which forces ALDI to build custom connections for every new project as seen with the ESL project, where the team developed a special "software component that migrates data from our ALDI systems to the manufacturer's cloud systems." This approach is inefficient and leads to project delays. It is also a main cause for operational errors, such as discrepancies between the online and in-store inventory or pricing.

Infrastructure Diagram



This is the ALDI infrastructure diagram we have created based on our research of what infrastructure ALDI currently employs.

It shows the 4 layers of the physical stores, and data centres, cloud services and customers as well as their connections to each layer above or below it.

The dashed "Complex Custom Integration" line shows the primary infrastructure gap in this analysis which is the fragile connection between legacy on premise systems and modern cloud services that create scalability, security and integration

challenges.

Figure 1: Infrastructure Analysis Diagram

B2C and B2B Analysis

E-commerce

B2C:

- As ALDI is mostly known for their presence physically with their supermarkets worldwide, there are cases where ALDI do reach their customers online.
- ALDI continues to engage with its customers online through their website and on their mobile app where they advertise new products and inform customers of seasonal promotions or any new deals currently happening.
- In the US, ALDI has partnerships with delivery apps such as Deliveroo, allowing customers to order and pay through the Deliveroo app to get their shopping directly delivered to their home.

B2B:

- In Ireland, ALDI uses EDI (Electronic Data Interchange) platforms to connect with its suppliers, meaning that the exchange of invoices, purchase orders and advanced shipping notices are in a structured electronic format. This allows ALDI to become more efficient with processing orders and reduce the number of errors. The EDI systems also allow ALDI to maintain its discount cost model as there is better inventory control and supply chain efficiency.
- ALDI Ireland runs a program called *Grow with ALDI* that supports Irish producers to become suppliers.

Custom Relationship Management (CRM)

B2C:

- ALDI uses their online platforms (website and mobile app) to send out notifications or emails to customers about products they frequently buy, or seasonal buys based on their shopping history.
- ALDI provides customer support where customers can submit their queries in which ALDI tracks it and responds efficiently, enhancing customer satisfaction.

B2B:

- ALDI usage of EDI also allows CRM to come into effect too. ALDI uses CRM to get information about their supplier (contracts and product lines), interactions (emails, phone calls, portal submissions) and performance (delivery reliability, order accuracy, compliance with ALDI's standards).
- CRM acts nearly as a hub for ALDI's supplier data, giving them a clear picture of their supplier's performance, history and status.

Application Programming Interface (API)

B2C:

- With the use of API, ALDI provides real time access to product catalogues, pricing, availability of a product and special offers. Their website and mobile app most likely use APIs to display up to date products and promotions to their customers.
- In other markets such as the US, since ALDI works with delivery platforms, API is used to send orders, retrieve status updates and synchronize inventory.

B2B:

- ALDI's EDI system integrates APIs to allow real time connection to their supply chain system meaning that supplies can send stock updates and other transactional data to ALDI via API which gets updated into ALDI's inventory management and supply chain analytics.

Data Sharing

B2C:

- ALDI collects data from their customers' interactions from their mobile app, visits on their websites and online orders (US market). This data would be presumably shared within ALDI to create personalized promotions, push notifications or email offers.
- ALDI's inventory management would also be presumably updated according to what their customers are buying as to prevent overselling.

B2B:

- ALDI shares purchase orders, stock levels and shipment confirmations with suppliers so as to avoid any penalties and stay in line with EDI compliance, to further back this it would be fair to assume that ALDI may also share delivery performance and demand of products back their suppliers through portals so suppliers can adjust production or delivery schedules.

Business Intelligence

Business Intelligence is important for Aldi to stay ahead of its competitors and decide what products should be ordered and the amount during certain times. Since Aldi doesn't track customers spending they have to base all their research from the sales history in the database. Let's explore how they can improve this and what's wrong currently

Data Sources

ALDI integrates data from a variety of internal and external sources to create a broad view of its operations and market:

Point of Sale (POS) Systems	ALDI's most critical internal source, every transaction captures granular data on movement of products, store location and final sale price.
Supply Chain & Inventory Management Systems	These systems track real time inventory levels, supplier delivery times, and product waste. This provides a unified view of costs and performance
Enterprise Resource Planning (ERP) Systems	ALDI uses a centralized SAP system. This integrates finance, logistics and human resource data. It provides a unified view of costs and performance

Analytic Tools

SAP Analytics Cloud	ALDI leverages the SAP ecosystem for enterprise reporting and planning. The SAP system enables analysis based on real time data.
SQL based Tools & Custom Applications	Core operational reporting uses custom applications. This ensures speed for daily sales and inventory restock.

Dashboards

Store Manager Dashboard	Shows a single store's performance: Daily sales vs target, out of stock alerts
Regional Manager Dashboard	Compares performance across a region Regional Sales, comparative store metrics, shrinkage rates
Category Buyer & Supply Chain Dashboard	Focuses on project movement Sales Velocity, Supplier Performance, Inventory Days of Supply

Key Performance Indicators (KPI)

ALDI's KPI's reflect its efficient operating model.

Sales per Square Foot	Measures store productivity
Inventory Turnover	Measures how many times a company sells and replaces its inventory over a given period
Gross Margin Return on Investments	Measures inventory profitability
Shrinkage as a % of Sales	Monitors waste from unsold goods, this protects margins and informs inventory ordering
Items per Customer Transaction	Tracks customer purchasing behaviour.

GDPR and Privacy**GDPR and Privacy: Aldi rewards app (IE)**

Creating a rewards scheme for Aldi would mean they would deal with a lot of personal data, that when they hold they have to follow certain guidelines. Customers must be fully aware of the data Aldi will take from them and they must have the customers full consent when they process this data. Since we plan the app to have features like personalized offers, digital coupons and rewards, we need to be careful. Since Aldi operates in Ireland, this means complying with the GDPR (General Data Protection Regulation), just like their current privacy notice says they do. We also plan to meet international standards like ISO 27001 and ISO 27701, which will help us build the app with privacy built in by design and by default.

Aldi's existing privacy notice shows they already take this seriously. For our app we will use that as a foundation and apply the same principles: being clear about what data we collect, only using it for the purposes we explain and keeping it secure. The main difference with our app is that we'll be using data more actively for personalization, so we need to be extra transparent about how that works.

Data Protection

For our Aldi rewards app, we'll implement strong data protection measures that align with GDPR requirements and Aldi's existing security standards.

We'll use end-to-end encryption for all customer data, specifically using the TLS 1.3 protocol to protect information while it's in transit, and AES-256 encryption for data while it's at rest. Access to this data will be restricted through role-based access control (RBAC) and multi-factor authentication ensuring that only authorised Aldi staff can process customer information.

To further protect privacy, transaction data will be masked through the application of loyalty points which reduces the risk of re-identification. All customer data will be hosted on EU-based cloud infrastructure authorized under ISO 27001 standards. This certification ensures regular privacy audits and ethical penetration testing to identify and address potential vulnerabilities within our databases.

We'll also maintain a formal data-breach response plan. In the event of any incident, we'll ensure reporting to supervisors occurs within 72 hours following GDPR Article 33 requirements.

User Consent

Following standard practice, our Aldi rewards app will include user consent through clear opt in procedures. Customers will have the choice to agree to share their data for marketing communications, app analytics and location-based offers. Their consent preferences will be stored securely and can be withdrawn at any time through the app's privacy settings.

The app will also request cookies and tracking permissions for browsing features like Aldi's current website approach. All privacy statements will be written in clear and concise language to ensure full transparency and fairness to customers making it easy for them to understand what they're agreeing to.

Lawful Basis

All ALDIs processing activities are related to the GDPR Article 6 of lawful bases:

- Contract Obligations - Managing user accounts, award points and award delivery
- Legitimate Interests - Maintaining app performances, fraud detection and analysing anonymous purchase trends.
- Consent - Personalised marketing and analytics.
- Legal Obligation - To retain transaction records for tax and accounting purposes

Each processing activity will be documented in Aldi's records of processing activities to ensure accountability and compliance.

Data Retention and Rights

For our Aldi rewards app, we'll retain personal data only while a customer's account shows active use, or longer if needed for official audits—after which it will be automatically deleted. Customers will have clear rights under GDPR Articles 15-20 to access, download, correct or delete their personal information. Any such requests will be handled through a formal workflow overseen by Aldi's Data Protection Officer (DPO), ensuring proper procedure and response.

Cross-Border Transfers and Governance

We will primarily store and process customer data within the EU, following ISO 27701 standards. However, some anonymised data may be shared for global analytics purposes. Any such transfers will be covered by standard contractual clauses (SCC) and managed through integrated automation systems (TIA) to ensure data remains always protected to EU privacy standards.

Conclusion

Aldi are serious about following guidelines so when it will finally come to implementing these guidelines to the final app, all the work will be there. They have to focus on the clear consent options, being very transparent about the use of data and follow all other GDPR guidelines to build the trust they need with customers.

Ethics Canvas

Figure 2 – Ethics Canvas for Aldi royalty scheme



Ethics Canvas Analysis

Introduction

The ethics canvas that we have provided in Figure 2 gives a rough overview on the ethical aspects of our proposal of the Aldi rewards app. Giving rewards for customer loyalty can provide very effective benefits in the long term, however, it's critical to note and understand the ethical challenges that the rewards app can deliver to ensure it is properly addressed and dealt with properly such as stakeholder impact, fairness, transparency and possible bias.

Stakeholder impact

The canvas shows that there is a wide range of stakeholders that can be impacted in different ways. Aldi customers are divided into two groups with people that show use of the Aldi rewards app and customers who do not have the app, showing the disparity in digital divide as one section clearly has better perks than the other in relation to purchasing power and potential deals. Aldi employees, particularly those in higher management face new responsibilities regarding the rewards app implementation such as handling data and implementing the system for the rewards app to work, which can result in a change in their respective roles. Suppliers within the "Grow with Aldi" program may find threats to their relationship with Aldi due to the potential impacts of product placement within the rewards app which could result in their products not being as widely recognised as any mainstream company. Competing retailers will have pressure with the release of the Aldi rewards app which will lead into a shift in the modern data collection practices.

Fairness and Bias

There is a potential for unfair outcomes to happen when it comes to the Aldi rewards app. The "Worldviews" section in the ethics canvas highlights the digital divide between customers without smartphones and customers with smartphones which shows the true disadvantages for customers without as they do not have digital literacy. This shows the bias against elderly shoppers and customers with a low income. The "Product or service failure" section outlines the potential issues arising including algorithmic errors which may show unfair personalised pricing and the "Group conflicts" section has suggested suppliers of popular products have an unfair advantage as opposed to the smaller localised companies within the "Grow with Aldi" when being recommended to customers in the rewards app. These points give an outline on how the system can be perceived as unequal between existing parties.

Transparency

Transparency is an essential factor which can result in trust issues across different stakeholders through a range of issues. The "Relations" section states that customer trust has a chance of being threatened if data is misused, which is a risk if data practices are not being followed. The safeguards that we provide in the Aldi rewards app, such as reports on how your data is used, are a direct solution to the problem. These reports are there to keep full transparency in check to not bewilder the customers on how their data is being used.

Resolution and Mitigation

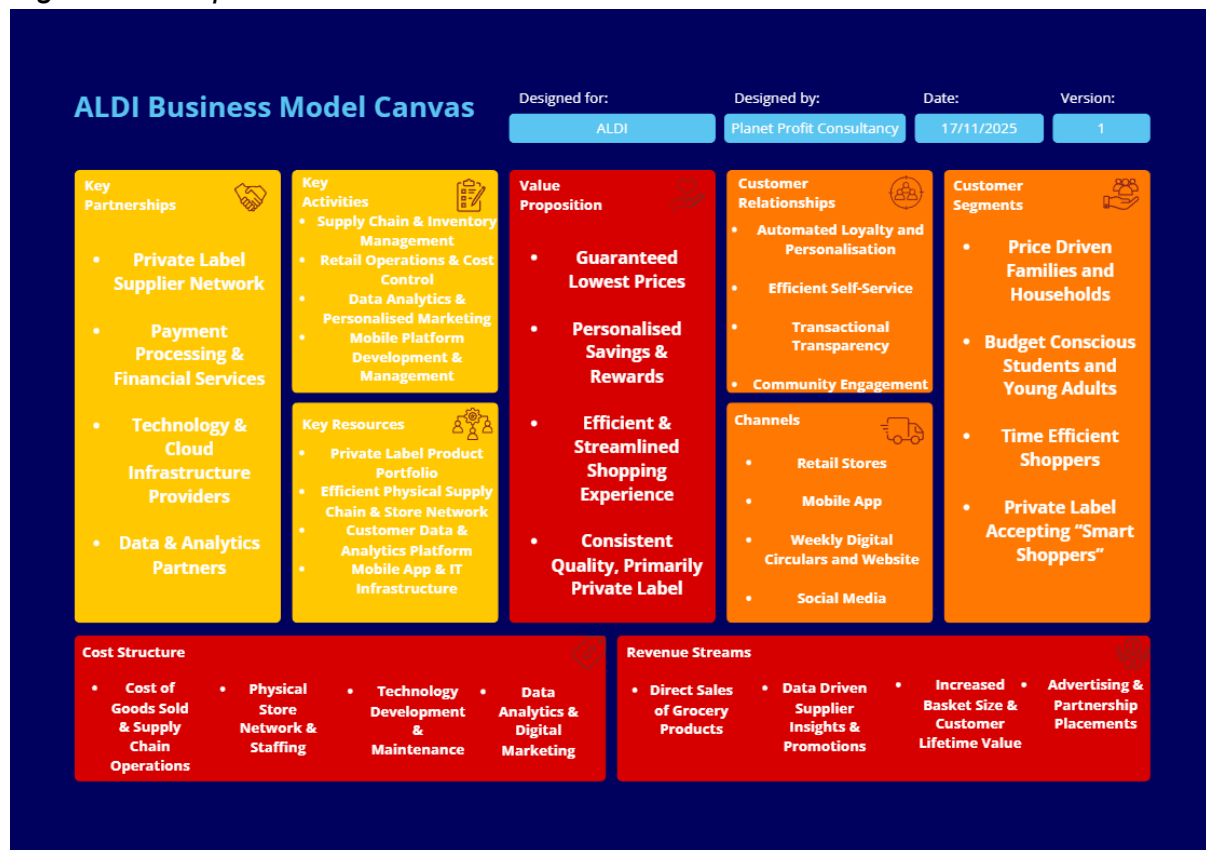
The "What can we do" section of the canvas shows us a fix for these problems. Using simple privacy controls, offering options that are not specific to the app, and being clear on how the app uses data, we can make the system go towards the right direction from the beginning. Putting ethics first in our approach is ensuring we make the app trustworthy and fair, transforming risks into a foundation for confidence in customers.

Conclusion

To conclude, this ethics canvas has shown that even through the rewards app offers a range of business intelligence tools and customer engagement opportunities, implementing the app ethically is non-negotiable. There is potential for biased algorithms, no transparency for data usage and stakeholder impacts which can be easily dealt with by upholding our proposed safeguarding strategies which ensures that the system builds on a good trustworthy relationship with the customer, maintaining equality on both sides.

Business Model Canvas

Figure 3 – Proposed Business Model Canvas for ALDI



This Business Model Canvas visualises our proposed plan for ALDI. We adopted a standard business model canvas that incorporates the specific changes driven by our new IS strategy.

Customer Segments

Our IS strategy allows Aldi to deepen its relationship with its core value seeking customers by moving beyond broad categorisation. The rewards app uses purchase data to deliver personalised value, whether that means offering targeted discounts to price driven families on their regular grocery staples, providing relevant budget deals to students, enhancing the efficiency for time poor shoppers with digital lists, or rewarding private-label advocates for their brand loyalty. This data driven approach ensures the value proposition feels uniquely relevant to each shopper.

Value Proposition

The IS strategy enhances Aldi's core value by adding a layer of personalisation to its famous efficiency and low prices. The ALDI Royalty Scheme transforms the promise of low prices into personalized savings, delivering tailored discounts and rewards directly to customers based on their actual shopping habits. This strengthens the value for all customers, from budget-conscious families seeking the lowest prices to time poor shoppers who value a streamlined experience by making the core offer of low prices more relevant and impactful for everyone.

Channels

Our IS strategy introduces the mobile application which transforms how value is delivered. While physical stores remain essential for product fulfilment, the app becomes the main channel for customer engagement, personalized communication, and loyalty management. It works alongside other channels like social media and digital circulars, creating a cohesive experience where marketing drives app downloads and the app in turn drives footfall to the physical stores.

Customer Relationships

Our IS strategy shifts Aldi's customer relationships from transactional and anonymous to personalized and data informed. The core of this new relationship is automated loyalty and personalisation managed through the rewards app. While Aldi will maintain its efficient self-service model in stores, the app creates a continuous and direct communication channel. This allows for personalised offers and engagement making customers feel individually valued and strengthening their connection to the Aldi brand beyond the checkout.

Revenue Streams

The IS strategy diversifies Aldi's revenue model by creating new, data driven income streams alongside core grocery sales. The primary new revenue comes from monetising aggregated customer data by selling insights and targeted promotion services to suppliers. Additionally, the app creates an advertising channel for partner brands. Furthermore, the loyalty program itself is designed to increase the main revenue stream by boosting average basket size and fostering customer loyalty leading to higher lifetime value.

Key Activities

The IS strategy we propose introduces new key activities centred around technology and data management. Alongside Aldi's established key activities of supply chain and retail operations, the company must now excel at mobile platform development and management to maintain the app. A new and equally important activity is data analytics and personalised marketing which involves processing customer data to generate insights and create targeted campaigns that drive the loyalty program.

Key Resources

The IS strategy we created establishes new key resources that are vital for the digital loyalty program. Alongside Aldi's foundational resources of its private label portfolio and efficient supply chain, the customer data and analytics platform become a critical asset. This data is essential for personalisation. Additionally, the mobile application and its supporting IT infrastructure become a key resource serving as the primary new channel for customer engagement and value delivery.

Key Partners

Our IS strategy requires Aldi to form new key partnerships to build and maintain the digital ecosystem. Beyond its essential network of private label suppliers, Aldi must now partner with technology and cloud infrastructure providers for the app's platform, payment processing services for secure transactions and data analytics firms to transform customer data into actionable insights. These partnerships provide the specialised capabilities needed to execute the strategy efficiently.

Cost Structure

Our IS strategy significantly alters Aldi's cost structure by introducing new investment areas. While costs of goods sold and physical store operations remain the largest expenses, new substantial costs are added for technology development, app maintenance and data analytics. These are primarily fixed costs for platform upkeep and variable costs for digital marketing and funding the personalised rewards, transforming some traditional marketing spend into more targeted and data driven expenses.

References

- Ephanov N, 2025 "One Of Aldi's Biggest Competitors Offers Loyalty Perks (Including Freebies And Coupons)" URL: <https://www.foodrepublic.com/1994750/aldi-biggest-competitor-lidl-perks-coupons/> Accessed: October 2025
- Katariya, J, 2025 "ALDI Business Model: A Complete and In-Depth Analysis" URL: <https://www.moontechnolabs.com/blog/aldi-business-model> Accessed: November 2025
- Graham, A. "Micro Data Grocery Analysis – The Day Aldi Came to Town" URL: <https://www.fabledata.com/the-value-of-micro-data-analysis-in-the-grocery-sector-the-day-aldi-came-to-town> Accessed: November 2025
- Bigora, P, 2024 "Aldi adding Instacart's omnichannel tech to all US stores" URL: <https://www.grocerydive.com/news/aldi-instacart-omnichannel-tech-expansion-ecommerce/723430/> Accessed: November 2025
- O'Sullivan S, 2025 "Aldi Ireland Opens Applications For Grow With Aldi 2025" URL: <https://www.checkout.ie/retail/aldi-ireland-opens-applications-for-grow-with-aldi-2025-215369> Accessed: November 2025
- Crossfire, "EDI Integration with Aldi" URL: <https://www.crossfireintegration.com/solutions/trading-partners/aldi/> Accessed: November 2025
- Crossfire, "EDI Compliance" URL: <https://www.crossfireintegration.com/solutions/trading-partners/aldi/> Accessed: November 2025
- ALDI DX. "Shopping without a till thanks to cameras and AI" URL: <https://it-jobs.aldi-sued.de/en/latest-insights/stories/shopgo> Accessed: November 2025
- ALDI DX. "ALDI DX. Electronic Shelf Labels: IT innovation in our store" URL: <https://it-jobs.aldi-sued.de/en/latest-insights/stories/electronic-shelf-labels> Accessed: November 2025
- ALDI DX. "ALDI DX. Interview with Kathrin, Manager Projects GenAI." URL: <https://it-jobs.aldi-sued.de/en/latest-insights/stories/manager-gen-ai-kathrin> Accessed: November 2025
- ALDI DX. "ALDI DX. How the 'Global Facilities Management' SaaS application facilitates work in our stores." URL: <https://it-jobs.aldi-sued.de/en/latest-insights/stories/saas-application-global-facilities-management> Accessed: November 2025

ALDI DX, "ALDI DX. How the 'Global Facilities Management' SaaS application facilitates work in our stores." URL: <https://it-jobs.aldi-sued.de/en/latest-insights/stories/saas-application-global-facilities-management> Accessed: November 2025

ALDI DX, "ALDI DX. Interview with Ievgenii, Security Architect." URL: <https://it-jobs.aldi-sued.de/en/your-possibilities/cloud-security-expert/testimonial-ievgenii-security-consultant> Accessed: November 2025

ALDI DX, "ALDI DX. Interview with Sabine, Solution Architect." URL: <https://it-jobs.aldi-sued.de/en/your-possibilities/developer/testimonial-developer-sabine> Accessed: November 2025

ALDI DX, "ALDI DX. Data is in our DNA: the ALDI SÜD universe now has its own Data & Analytics unit." URL: <https://it-jobs.aldi-sued.de/en/our-teams/data-analytics/data-analytics-unit-aldi-south> Accessed: November 2025

ALDI DX, "ALDI DX. Interview with Roland, Product Owner." URL: <https://it-jobs.aldi-sued.de/en/your-possibilities/agile-expert/testimonial-product-owner-roland> Accessed: November 2025

ALDI DX, "Business Strategy Hub. ALDI Business Model | How Does ALDI make money?" URL: <https://bstrategyhub.com/aldi-business-model-how-does-aldi-make-money/> Accessed: November 2025

ALDI STORES, ALDI STORES (IRELAND) LIMITED CUSTOMER PRIVACY NOTICE <https://www.aldi.ie/privacy> Accessed: November 2025

LIDL, Data Protection Information My LIDL Account <https://www.lidl.ie/c/privacy/s10024085> Accessed: November 2025

The Business Model Analyst. ALDI Business Model URL: <https://businessmodelanalyst.com/aldi-business-model/> Accessed: November 2025

Fox, G. "ALDI Business Model: Unveiling the Secrets of Its Success" URL: <https://www.garyfox.co/aldi-business-model/> Accessed: November 2025

ALDI DX, "Data & Analytics @ ALDI SÜD" URL: <https://it-jobs.aldi-sued.de/en/our-teams/data-analytics/data-analytics-unit-aldi-south> Accessed: November 2025